

Managing Research Projects

| A practical system for getting research finished

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Every lab has a graveyard of projects that have been 80% done for five years.

Projects fail from entropy, not stupidity

- Undefined endpoints — nobody said what “done” is
- Mission creep — the question moved after the data arrived
- Entropy — files, decisions, and intentions decay

EVERYTHING TODAY IS A CHEAP, MECHANICAL DEFENSE AGAINST ALL THREE FAILURE MODES.

Roadmap

1. What a project is (and isn't)
2. Choosing the project
3. The protocol — why written plans change outcomes
4. Structure: repositories, folders, reproducible code
5. Git and GitHub, the concept
6. Driving to done
7. **You draft a protocol** + discussion

What a project is

What are you working on?

A project is the work aimed at one product

- **Project** = all the documents, data, and code aimed at one **product**
- **Product** = something you can deliver: a paper, a chapter, a package, a data archive
- One project = one product = one repository

If you can't write a plan for it, it isn't a project

- A dissertation is not a project; **each chapter is**
- Peer reviews, proposals, and presentations are not projects
- An NSF or NIH grant is not a project either — it's an **umbrella funding mechanism** that pays for several
- A project is *over* when the product is delivered — the second paper from the same data is a **new project**, not scope growth

Choosing the project

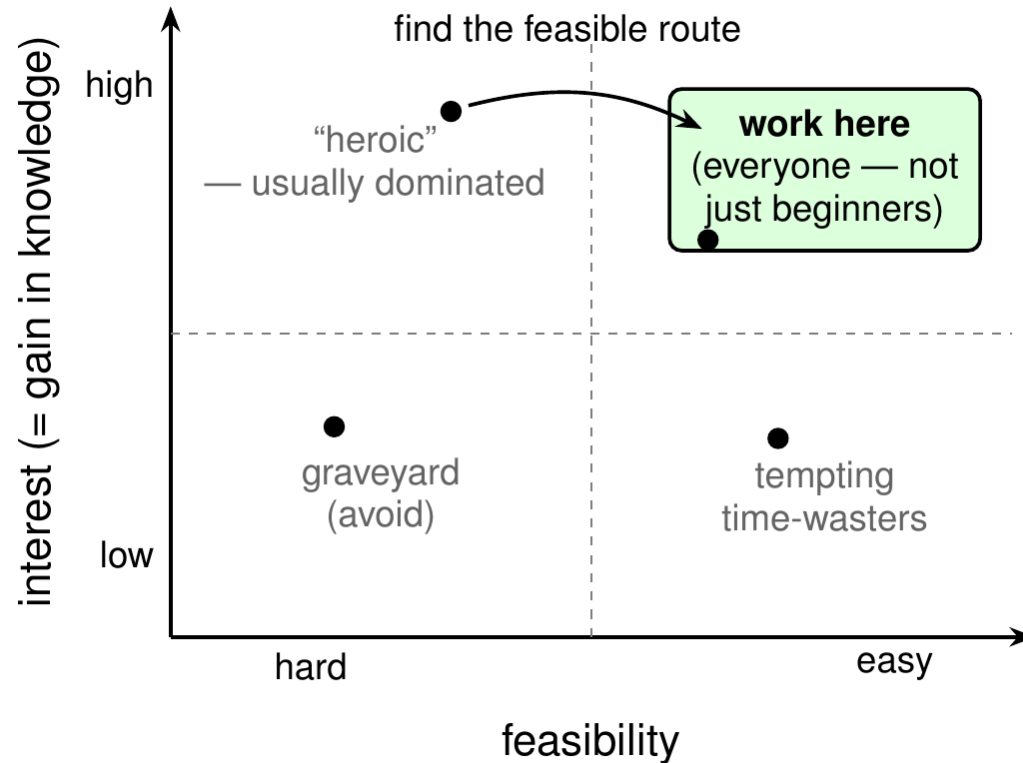
The highest-leverage decision you will make

The stakes: only about 57% of PhD students finish within ten years

- ~57% ten-year completion (~63% in the life sciences), students entering 1992–95 at ~30 universities
- Project choice is the highest-leverage, least-taught decision in a research career
- Time spent choosing — weeks, even months — is not procrastination

Source: Council of Graduate Schools, PhD Completion Project (2008). Alon (2009): "Do not commit to a problem before 3 months have elapsed."

Plot every candidate project on two axes



“If you do not work on an important problem, it’s unlikely you’ll do important work.”

*— Richard Hamming,
“You and Your Research”
(1986)*

Candidate projects plotted by feasibility and interest (gain in knowledge); adapted from the same two axes in Fig. 1 of Alon (2009), *Mol Cell* 35:726.

Where I part ways with Alon: everyone works easy/high-gain

- Alon: first problems are easy and modest; postdocs need feasible *and* high-gain; new PIs may take on a hard grand challenge — if it splits into tractable projects
- My advice: **difficulty is not a virtue — it is a cost that discounts everything downstream**
- For almost every hard problem there is an easier project that teaches the same lesson — **find the easier route before you commit**

Most published science is conservative — for a reason

- Across ~6.5 million biomedical abstracts, **86% of the chemical relationships reported were already known**; 14% were new
- Papers introducing a new chemical were cited more — **12.9 citations on average versus 8.4** — but a project that fails produces no paper at all, and by the authors' calculation the extra citations don't make up for that risk
- Highly novel papers had **~57% higher odds of a top-1% hit** — and fatter tails at both ends
- In a field experiment, expert reviewers gave the most novel proposals markedly **lower** scores

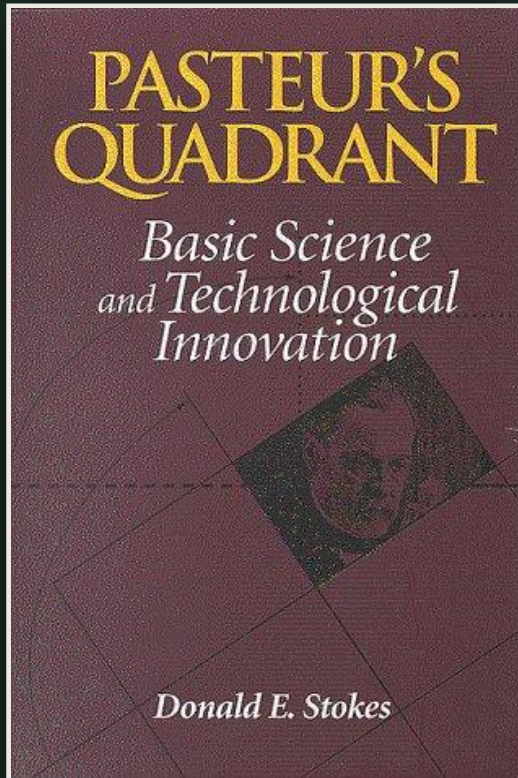
Sources: Foster, Rzhetsky & Evans (2015) *Am Sociol Rev* 80:875; Wang, Veugelers & Stephan (2017) *Res Policy* 46:1416; Boudreau et al. (2016) *Manag Sci* 62:2765.

The highest-impact papers are conventional work with one unusual combination

- 17.9 million papers, scored by how usual or unusual each pair of cited journals is
- Papers that were mostly conventional but included **one rare combination** reached the top 5% of citations **9.1% of the time** — versus 5.8% for purely conventional papers and 5.3% for purely novel ones
- Only **7%** of papers achieve that combination
- Teams of three or more included an unusual combination in **half** their papers; solo authors in a third

Source: Uzzi, Mukherjee, Stringer & Jones (2013) *Science* 342:468.

Donald Stokes and Pasteur's Quadrant



- Donald E. Stokes (1927–1997) — political scientist; co-author of *The American Voter* (1960); dean of Princeton's Woodrow Wilson School, 1974–1992
- *Pasteur's Quadrant* (Brookings, 1997) was published shortly after his death
- His target: the old **linear model** — basic research upstream, applied downstream, purity and usefulness as opposite ends of one axis
- Stokes split that into **two independent questions** — and the quadrant falls out

Point the project with Pasteur

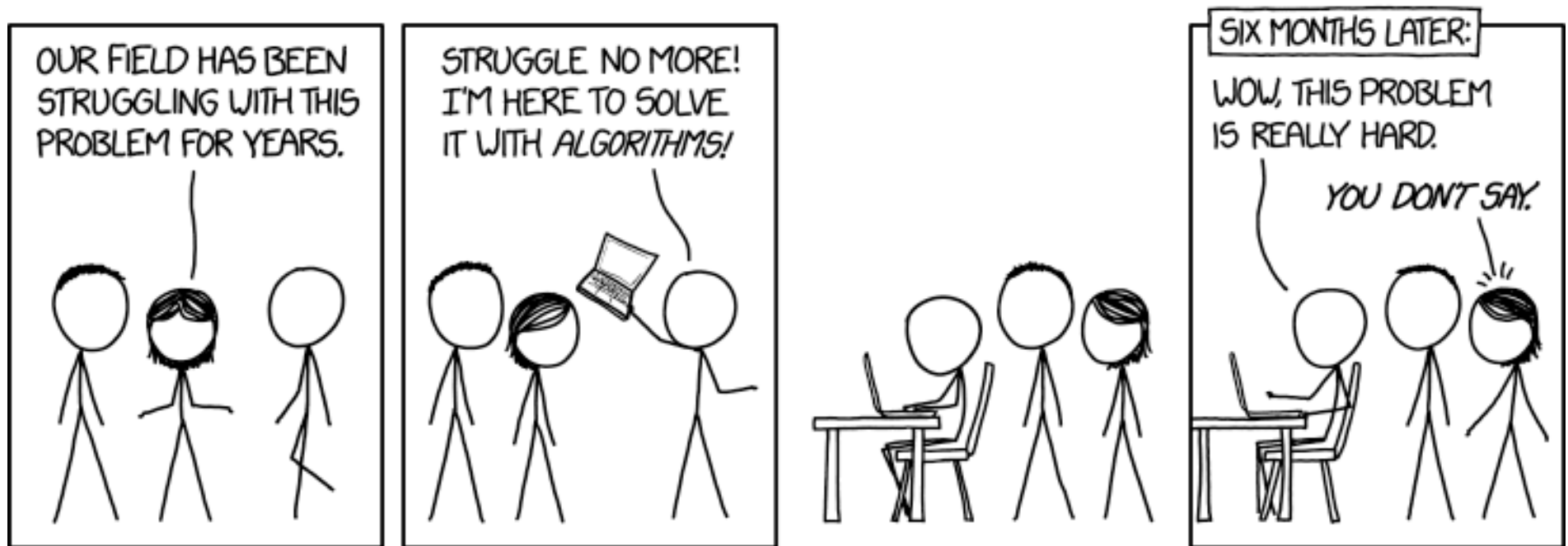
		Inspired by considerations of use?	
		no	yes
Seeks fundamental understanding?	yes	Bohr pure basic research (atomic structure)	Pasteur use-inspired basic research (disease ecology lives here)
	no	(Peterson's field guides)	Edison pure applied research (the light bulb)

Stokes (1997): two axes, not one. Use-inspired basic research — Pasteur's quadrant — is where understanding and consequence advance together.

Discussion

WHERE DOES YOUR CURRENT PROJECT SIT ON THE FEASIBILITY-INTEREST DIAGRAM?

COULD AN EASIER PROJECT TEACH THE SAME LESSON?



"Here to Help" — Randall Munroe, xkcd.com (CC BY-NC 2.5)

The protocol

Write the plan before the data exist

After trial registration arrived, positive results fell from 57% to 8%

- 55 large NHLBI trials of drugs and supplements, all with cardiovascular outcomes
- Published before 2000: **17 of 30 positive** on their primary outcome
- Published 2000 or later, with outcomes registered in advance on ClinicalTrials.gov: **2 of 25 positive**

Source: Kaplan & Irvin (2015) *PLoS ONE* 10:e0132382.

Four innocuous choices, left open, turn 5% false positives into 61%

- Two outcome variables instead of one; peeking before adding observations; an optional covariate; dropping a condition
- Each inflates error modestly; **combined: a 60.7% false-positive rate** at nominal $p < .05$
- No fraud required — it's enough that you *would have* analyzed different data differently (the “garden of forking paths”)

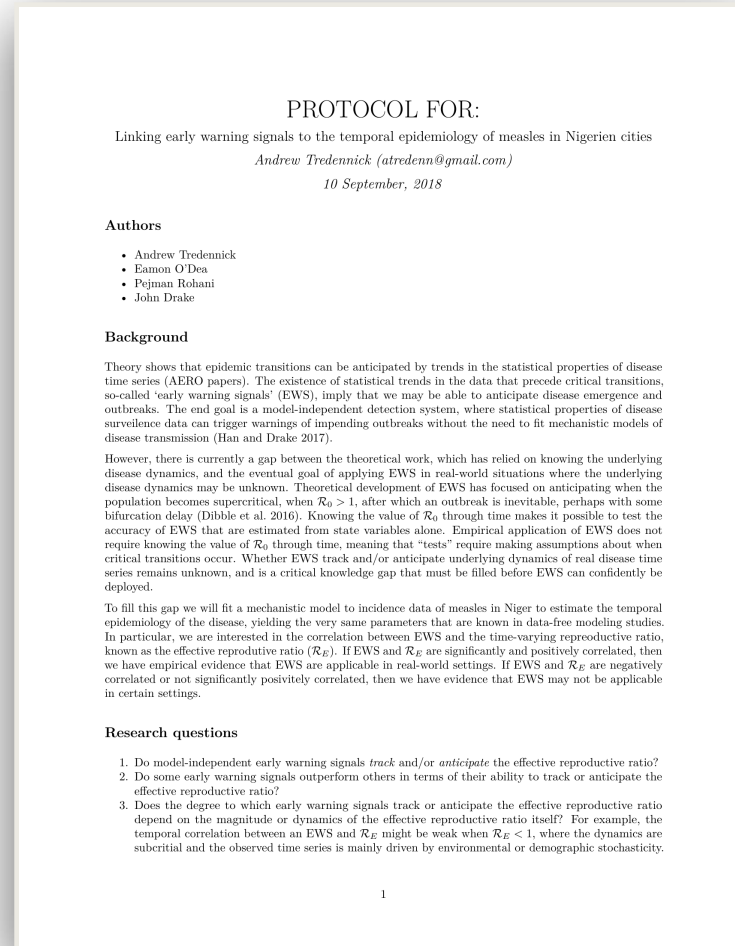
Sources: Simmons, Nelson & Simonsohn (2011) *Psych Sci* 22:1359; Gelman & Loken (2014) *Am Sci* 102:460.

This is us: 174 ecology teams, same data, opposite conclusions

- Two unpublished datasets (blue tit sibling competition; *Eucalyptus* recruitment) analyzed independently by 174 teams
- Effects ranged from significantly negative to significantly positive
- Peer-rated analysis quality **did not predict** who landed where
- In a survey of ecologists and evolutionary biologists, 64% admitted cherry-picking and 51% **HARKing** — Hypothesizing After the Results are Known: presenting an unexpected finding as if it had been predicted from the start

Sources: Gould et al. (2025) *BMC Biology* 23:35; Fraser et al. (2018) *PLoS ONE* 13:e0200303.

The Drake Lab protocol: a contract with your future self



A real protocol: Tredennick's 2018 measles early-warning study — background, research questions, and study design, set down at the outset and kept current as the work moved.

Seven parts, one page is enough to start

1. Title and researchers — an **originator** with contact info
2. Background and rationale — 2–3 succinct paragraphs
3. Study design — steps + explicit research questions (*for models: design it like an experiment — Drake 2025*)
4. Data sources — with permissions needed
5. Analysis — including the **anticipated figures**, sketched before the data exist (*for models: treatments, levels, and responses — Drake 2025*)
6. Checklist — tasks with dates and owners
7. Changelog — the plan may change, **but only in writing, with the reason**

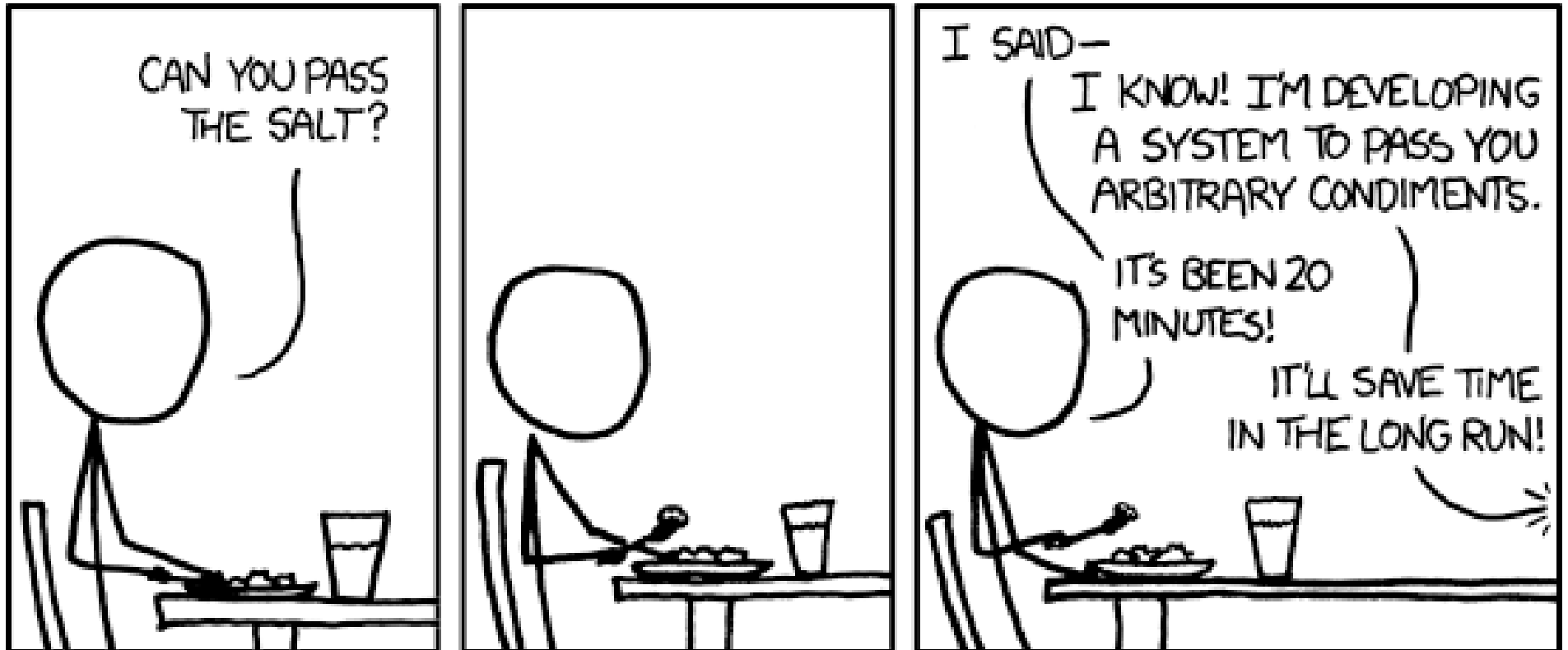
For computational projects, the study design is an experimental design

- **Treatments** — what you vary (parameters, scenarios, interventions)
- **Levels** — the values each treatment takes, chosen in advance
- **Responses** — the outputs you will measure, named before the first run
- **Replication** — how many stochastic realizations, and the seed policy
- **Control** — the baseline scenario everything is compared against

After Drake (2025), "Modelling Like an Experimentalist," *Ecology Letters* 28:e70251.

Discussion

IF YOUR CURRENT PROJECT HAD A CHANGELOG, WHAT WOULD ITS FIRST ENTRY SAY?



"The General Problem" — Randall Munroe, xkcd.com (CC BY-NC 2.5)

Structure

Repositories, folders, and code that reruns

One project = one repository, named so a stranger can find it

- Create the repository **the day the protocol exists**
- Naming: **surname-description** (individual) or **drakelab-description** (team)
- **README.md** is the front door: project name, originator, collaborators, the protocol — and a **map of how the repository is organized**
- Private project, public product: the repo is the full internal history; the product (paper + archive) is the curated public face

A standard layout means never asking “where does this go?”

```
project/
├── README.md          protocol, or pointer to it
├── data/              raw, read-only, never edited
├── scripts/           01-clean.R, 02-fit.R, 03-figures.R
├── output/            derived data + figures, regenerable
└── documents/         manuscript, notes, correspondence
```

TWO RULES DO MOST OF THE WORK: RAW DATA ARE SACRED; EVERYTHING DERIVED IS DISPOSABLE.

After Noble (2009) *PLoS Comput Biol* 5:e1000424 and Wilson et al. (2017) *PLoS Comput Biol* 13:e1005510.

Undocumented projects decay

- **Information entropy**: detail is lost from the moment of publication — specifics first, then generalities, with step losses at retirement, accidents, and death
- Odds a dataset still exists fall **17% per year** after publication
- Even under strong archiving policies, **56%** of ecology datasets were incomplete and **64%** archived in ways that partly or wholly prevented reuse

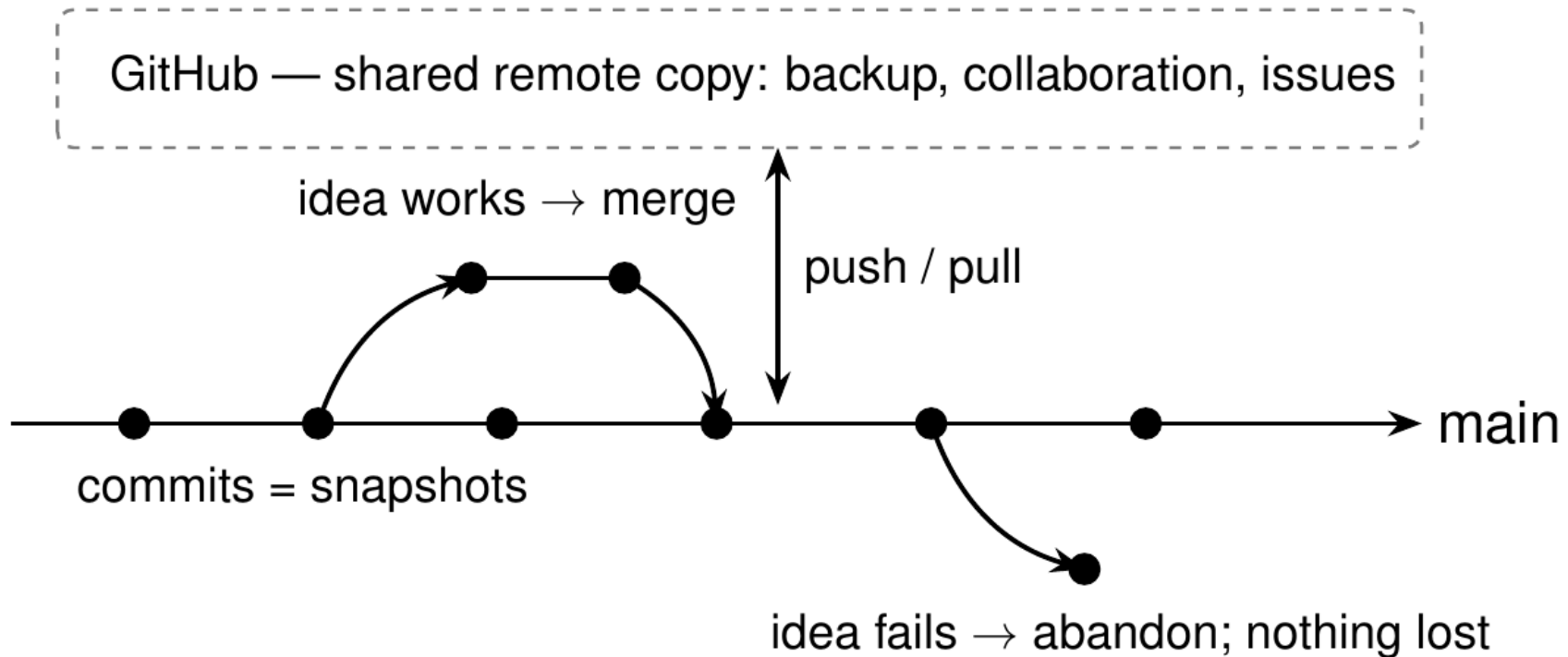
Sources: Michener et al. (1997) *Ecol Appl* 7:330; Vines et al. (2014) *Curr Biol* 24:94; Roche et al. (2015) *PLoS Biol* 13:e1002295.

*The test: if **output/** was lost tonight, could one command rebuild it?*

- When published R code was re-run in a clean environment: **74% of scripts failed** — library and working-directory errors among the most common
- In ecology journals with code-sharing policies, 79% of articles shared data but **only 27% shared code**
- Everything by script; relative paths; record seeds and package versions

Sources: Trisovic et al. (2022) *Sci Data* 9:60; Culina et al. (2020) *PLoS Biol* 18:e3000763; Sandve et al. (2013) *PLoS Comput Biol* 9:e1003285; Noble (2009).

Version control is a lab notebook you cannot lose



COMMIT PER LOGICAL CHANGE; PUSH AT LEAST ONCE A DAY.

Commits are dated snapshots of the whole project; branches let ideas fail safely; GitHub holds the shared copy. Mechanics — branching, merging, pull requests — get their own hands-on workshop: **October 20, 2:55 PM**.

Driving to done

The checklist is the engine; dependencies are where projects stall

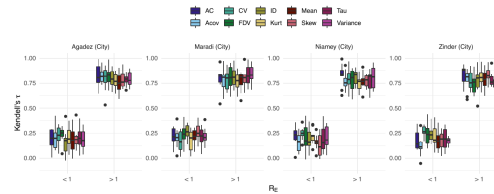


Figure 4: Boxplots of temporal correlations between EWS and R_E through time, split apart by the value of R_E .

Checklist

- Create cleaned dataset of just the four focal cities from Niger
- Write R code to define the pomp model object (SDE)
- Simulate time series from the pomp object with parameters that give us irregular dynamics like the observed data
- Do test analysis of steps II-III using the simulated data
- Perform iterated filter parameter search for MLEs
 - Get set up on UGA cluster
 - Get set up on MIDAS cluster
- Calculate EWS on the observed data
 - Work with Eamon on bandwidth, etc.
- Calculate correlations with real data and empirically-derived R_E and lags
- Write the paper!

Changelog

- 09-10-2018
 - We are going to forego the proposed initial search via particle filtering because doing so over a large enough parameter space to be meaningful is too computationally expensive. Instead, after conferring with Pej, the new plan is to start with MIF from 1,000 starting parameter sets from a Latin Hypercube design. Then we'll gradually take subsets of those.
 - Imported infections are now stochastic: `ps1[t] ~ rpois(psi1)`.
 - Likelihood is now poisson. Initial fitting kept showing very little overdispersion, so we're going to use Poisson to speed up computation.

The Tredennick protocol's checklist, mid-project: struck-through tasks, surviving blockers, and a changelog recording why the plan changed.

Done means delivered — then you stop

- A project is over when the product is delivered: paper out, code and data archived (e.g., Dryad) with the repository linking to them
- New ideas from old projects become **spinoffs** with new protocols and new repos
- The portfolio view: one fast/safe, one core, one stretch — chapters as separate projects makes this natural

Exercise

Start your own protocol — right now

Fifteen minutes: draft the skeleton for one real project

1. **Name the product** — one sentence: what do you deliver?
2. **Background** — two sentences
3. **Research question** — and what would an answer look like? (a p-value? a pattern in a graph? a table of numbers?)
4. **Three checklist items** — each with a date
5. **One dependency** — the thing most likely to block you

THEN TWO MINUTES: TELL YOUR NEIGHBOR YOUR PRODUCT IN ONE SENTENCE.

Discussion

WHAT DID YOU DISCOVER DRAFTING IT? WHERE DO YOU DISAGREE WITH THE SYSTEM?

HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC)



"Standards" — Randall Munroe, xkcd.com (CC BY-NC 2.5)

The starter kit

- **Protocol template** + worked example: github.com/DrakeLab/Wiki → lab-docs → project-protocols
- **Folder skeleton**: README + data + scripts + output + documents
- **One repo per project**, created the day the protocol exists
- **Definition of done**: the product is delivered, the archive is public, you stop

YOU DON'T NEED MORE DISCIPLINE OR BETTER SOFTWARE. YOU NEED A PLAN YOU CAN PRINT, A REPO WITH FIVE FOLDERS, AND A DEFINITION OF DONE. WRITE THE PROTOCOL THIS WEEK.

Backup

Does better documentation of analytic choices fix the many-analysts problem?

- 161 researchers, 73 teams, one dataset (ISSP) and one hypothesis — does immigration erode support for social policy? — 1,261 models submitted
- Of 89 team-level conclusions: **~61% reject, ~26% support, ~13% “not testable”**; estimates ran from large negative to large positive
- After coding 107 analytic decisions plus expertise and prior beliefs, **95.2% of the variance in results remained unexplained**
- Expertise didn't shrink it, and no team changed its model after seeing the others' — the divergence is idiosyncratic, not bias

Source: Breznau et al. (2022) *PNAS* 119:e2203150119.

What do honest base rates look like?

- Standard psychology literature: **96% of first hypotheses “confirmed”** (n=152 papers)
- Registered Reports: **44%** (n=71)
- Across 113 Registered Reports in biomedicine and psychology, ~61% of 296 hypotheses were *not* supported

Sources: Scheel, Schijen & Lakens (2021) *AMPPS* 4(2); Allen & Mehler (2019) *PLoS Biol* 17:e3000246.

How much do effects shrink on replication?

- When an experiment is repeated, the measured effect is almost always smaller than the original report — the gap is the shrinkage
- Reproducibility Project: Cancer Biology — for positive effects, median replication effect **85% smaller** than original; **46%** of effects replicated by a majority of five criteria
- The project planned 193 experiments and completed 50 — missing data, under-described methods, and unhelpful authors blocked the rest
- Nature survey of 1,576 researchers: >70% failed to reproduce another's experiment; >50% failed to reproduce **their own**

Sources: Errington et al. (2021) *eLife* 10:e71601 and 10:e67995; Baker (2016) *Nature* 533:452.

Why doesn't the system just reward good methods?

- Model: when institutions reward publication volume, low-effort methods out-reproduce careful ones — **no misconduct required**
- Mean statistical power to detect small effects in the social and behavioral sciences: **0.24, unchanged 1960–2011**

Source: Smaldino & McElreath (2016) *R Soc Open Sci* 3:160384.

Doesn't “work on important problems” contradict easy/high-gain?

- “If you do not work on an important problem, it's unlikely you'll do important work.”
- But: “It's not the consequence that makes a problem important, it is that you have a **reasonable attack**.”
- Great scientists keep 10–20 important problems waiting for an attack to appear

Source: Hamming (1986), “You and Your Research,” Bell Communications Research Colloquium.

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